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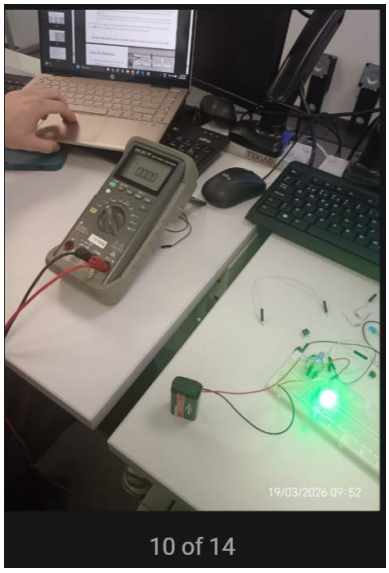
IT23SP

Analog Electronics 2

Circuit Challenge 5



The Night Light Company

Circuit Challenge 5: The Night Light Company
The final "million-dollar" challenge is to build an automated **Night Light**. The circuit must be "light-sensitive," meaning it remains off in a bright room and automatically illuminates the LED when the ambient light levels drop.

The "Millionaire" Design Strategy
To achieve automation, we move beyond manual push buttons and use a **Transistor** as an electronic switch, controlled by a **Voltage Divider**.

1. The Brain: The Voltage Divider
You will create a "sensor" by pairing your **LDR (Light Dependent Resistor)** with a standard resistor in series.

- **In the Light:** The LDR has very low resistance. Most of the voltage stays "before" the transistor, keeping it off.
- **In the Dark:** The LDR's resistance spikes (often to over 1MΩ). This causes the voltage at the "output" of the divider to rise.

2. The Switch: The NPN Transistor
The output from your voltage divider connects to the **Base** of the transistor.

- Once the voltage at the Base reaches approximately **0.7V**, the transistor "closes" the circuit.
- This allows current to flow from the **Collector** to the **Emitter**, completing the path for your LED.

3. The Output: LED & 200Ω Resistor

Just like in previous challenges, the LED is protected by its **200 Ω resistor**. This ensures that once the transistor flips the "on" switch, the LED shines at that high brightness we saw in Challenge 1 without burning out.

Challenge 6..... not done!

Challenge 7..... not done!

Build Together 1

Ping Pong Flashers

Circuit Report: Ping Pong Flashers (Astable Multivibrator)

Objective

The final project moves from manual and sensor-based switching to **automated oscillation**. The "Ping Pong Flasher" is an astable multivibrator circuit designed to alternate power between two LEDs (Red and Green) automatically, creating a rhythmic flashing effect.

How the "Ping Pong" Effect Works

This circuit relies on the interaction between **capacitors, resistors,** and **transistors** to create a feedback loop that never settles into a steady state.

- **The Capacitors ($100\mu F$):** These act as temporary energy reservoirs. They charge up through the resistors and then discharge into the base of the *opposite* transistor.
- **The Cross-Coupled Design:** * The cathode (short leg) of the **first capacitor** is connected to the **base of the second transistor**.
  - The cathode of the **second capacitor** is connected to the **base of the first transistor**.
- **The Alternating Switch:** When one capacitor discharges, it "turns on" the opposite transistor, which illuminates its LED and simultaneously begins charging the next capacitor. This "handoff" continues indefinitely as long as power is supplied.

Technical Observations

- **Timing Control (RC Time Constant):** The speed of the "ping pong" flash is determined by the **RC circuit** (the $10k\Omega$ resistors and the $100\mu F$ capacitors).
  - *Higher resistance/capacitance* = Slower flashes.
  - *Lower resistance/capacitance* = Faster flashes.
- **Component Safety:** Each LED remains protected by a **200 Ω resistor** connected to the collector of its respective NPN transistor, maintaining the high-brightness standard established in Challenge 1.